

Three Kinds of Evidence About a Limit

Answer Key

AP Calculus AB/BC Limits and continuity

Quick Answers

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|---------------------------------------|-------------------|
| 1. 8 | 6. 3 |
| 2. 5.9, 5.999, 6.001, 6.1, 6 | 7. $-\frac{1}{4}$ |
| 3. 2 | 8. -4, -2 |
| 4. $\frac{4}{3}$ | 9. -1, -1, 1, 1 |
| 5. 11.41, 11.9401, 12.0601, 12.61, 12 | 10. 4.9, 5.1, 5 |
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Curriculum

Level: AP Calculus AB/BC Limits and continuityLIM-1, LIM-2 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10**Difficulty 1–5 of 5 across 28 tagged checks.*

1. $\lim_{x \rightarrow 3} (x^2 - 2x + 5)$.

Step 1. Substitution is only legal when the function is continuous there, so the first move is to notice that this is a polynomial — continuous everywhere.

Step 2. Substituting: $3^2 - 2(3) + 5 = 9 - 6 + 5$.

Step 3. The limit is 8.

Why substitution is allowed: for a continuous function the limit at a point equals the value there. That sentence is the whole justification.

2. Table for $f(x) = \frac{x^2 - 9}{x - 3}$ near $x = 3$.

Step 1. The expression is undefined at 3, so no single substitution answers the question; the table has to approach from both sides.

Step 2. For $x \neq 3$ the fraction cancels to $x + 3$, which is what makes the cells easy to check.

x	2.9	2.999	3.001	3.1
$f(x)$	5.9	5.999	6.001	6.1

Step 3. The values close in on the same number from both sides, so $\lim_{x \rightarrow 3} f(x) =$ 6.

Teaching note: the table never reaches 6 — and it does not have to. A limit is what the values head toward, not a value the function attains.

3. The graph of g at $x = 1$.

Step 1. Limit and value are separate questions, so read the curve first and the dot second.

Step 2. Coming in from the left along $y = x + 1$ the curve heads for height 2; coming in from the right along the constant branch it is already at height 2. Both sides agree, so $\lim_{x \rightarrow 1} g(x) = \boxed{2}$.

Step 3. The filled dot sits at $(1, 4)$, so $g(1) = 4$. The limit exists but does not equal the value, which is a **removable** discontinuity — redefining $g(1) = 2$ would repair it.

Step 4 (algebraic confirmation). On the left branch $g(x) = x + 1$, so $\lim_{x \rightarrow 1^-} g(x) = 1 + 1 = 2$, matching the graph.

Listen for: “the limit is 4 because that is the point.” The open circle, not the filled dot, is where the curve is going.

4. $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - x - 2}$.

Step 1. Substitution gives $\frac{0}{0}$, which is not a value but a signal that numerator and denominator share the factor $x - 2$.

Step 2. Factor both: $\frac{(x - 2)(x + 2)}{(x - 2)(x + 1)}$. For $x \neq 2$ the common factor cancels, leaving $\frac{x + 2}{x + 1}$.

Step 3. That expression is continuous at 2, so substitute: $\frac{4}{3}$, giving $\boxed{\frac{4}{3}}$.

Common wrong answer: “does not exist” — $\frac{0}{0}$ was read as a verdict instead of as an instruction to factor.

5. Table for $f(x) = \frac{x^3 - 8}{x - 2}$ near $x = 2$.

Step 1. Again $\frac{0}{0}$ at $x = 2$, so build the numerical evidence first and then settle it with algebra.

Step 2. The difference of cubes factors as $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$, so for $x \neq 2$ the function is $x^2 + 2x + 4$.

x	1.9	1.99	2.01	2.1
$f(x)$	$\boxed{11.41}$	$\boxed{11.9401}$	$\boxed{12.0601}$	$\boxed{12.61}$

Step 3. The cells bracket the same number from both sides, and substituting into $x^2 + 2x + 4$ confirms it: $4 + 4 + 4$, so the limit is $\boxed{12}$.

Teaching note: this is the pattern to internalise — the table says *about* 12, the algebra says *exactly* 12.

6. The graph of g at $x = 3$.

Step 1. When the two sides of a graph arrive at different heights the one-sided limits must be read separately.

Step 2. From the left the constant branch is at height 2, so $\lim_{x \rightarrow 3^-} g(x) = 2$. From the right the falling branch starts at the open circle at height 3, so $\lim_{x \rightarrow 3^+} g(x) = 3$.

Step 3. A two-sided limit exists only when the one-sided limits are equal. Here $2 \neq 3$, so $\lim_{x \rightarrow 3} g(x)$ does not exist — a **jump** discontinuity.

Step 4 (algebraic confirmation). On the right branch $g(x) = 6 - x$, so $\lim_{x \rightarrow 3^+} g(x) = 6 - 3$, which is $\boxed{3}$ as the graph showed.

Note: $g(3) = 2$ is defined — a jump discontinuity does not require a hole in the domain.

7. $\lim_{x \rightarrow 0} \frac{\frac{1}{x+2} - \frac{1}{2}}{x}$.

Step 1. The compound fraction hides the cancelling factor, so combine the numerator over a common denominator before touching the outer division.

Step 2. $\frac{1}{x+2} - \frac{1}{2} = \frac{2 - (x+2)}{2(x+2)} = \frac{-x}{2(x+2)}.$

Step 3. Dividing by x means multiplying by $\frac{1}{x}$: $\frac{-x}{2x(x+2)} = \frac{-1}{2(x+2)}$ for $x \neq 0$.

Step 4. That is continuous at 0, so substitute: $\frac{-1}{2(2)}$, giving $\boxed{-\frac{1}{4}}$.

Common wrong answer: $-\frac{1}{2}$ — the x in the outer denominator was cancelled twice.

8. Choosing a so that $f(x) = \frac{x^2 + ax + 3}{x - 1}$ has a finite limit at 1.

Step 1 (a). The denominator goes to 0 at $x = 1$. A quotient with a nonzero numerator over 0 blows up, so a finite limit forces the numerator to vanish at 1 as well.

Step 2 (a). Setting the numerator to zero at $x = 1$: $1 + a + 3 = 0$, so $\boxed{a = -4}$.

Step 3 (b). With $a = -4$ the numerator is $x^2 - 4x + 3 = (x - 1)(x - 3)$, so for $x \neq 1$ the function equals $x - 3$.

Step 4 (b). Substituting, $1 - 3$, so $\lim_{x \rightarrow 1} f(x) = \boxed{-2}$.

Check: any other a leaves a nonzero constant over a vanishing denominator, and the one-sided limits run off to $+\infty$ and $-\infty$.

9. Table for $f(x) = \frac{|x - 4|}{x - 4}$ **near** $x = 4$.

Step 1. Absolute value is a two-case rule, so the sign of $x - 4$ has to be tracked separately on each side.

Step 2. For $x < 4$, $|x - 4| = -(x - 4)$ and the quotient is -1 ; for $x > 4$, $|x - 4| = x - 4$ and the quotient is 1 .

x	3.9	3.99	4.01	4.1
$f(x)$	-1	-1	1	1

Step 3. Both one-sided limits exist — -1 on the left, 1 on the right — but a two-sided limit requires them to be equal, and they are not. So $\lim_{x \rightarrow 4} f(x)$ does not exist.

Grading note (manual): full credit needs the reason, not just “DNE”: the values do not settle on one number because the two sides settle on different ones.

10. Synthesis: $p(x) = \frac{x^2 + x - 6}{x - 2}$.

Step 1 (a). For $x \neq 2$ the numerator factors as $(x - 2)(x + 3)$, so $p(x) = x + 3$ everywhere except at 2 — which is what makes the table cells one-step evaluations.

Step 2 (a). $p(1.9) = \boxed{4.9}$ and $p(2.1) = \boxed{5.1}$.

Step 3 (b). Since $p(x) = x + 3$ near 2 , $\lim_{x \rightarrow 2} p(x) = 2 + 3 = \boxed{5}$. Defining $p(2) = 5$ removes the discontinuity, because then the value equals the limit.

Step 4 (c). The sketch is the line $y = x + 3$ with an open circle at $(2, 5)$ — one point punched out of a straight line, nothing else.

Grading note (manual): the written explanation should say that the table *suggests* 5 , the algebra *proves* it, and the sketch *shows* why the function still fails to be continuous until $p(2)$ is defined to be that number. Any two of the three without the third is a partial answer.